Cormalent Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-766/C

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. The control strategy links critical quality attributes to critical process parameters identified during development. Deviations observed during the campaign were investigated and closed with no impact to product quality. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Control of Excipients (3.2.P.4)

For the commercial formulation F-766/C, Aerosil 200 Pharma (Colloidal silicon dioxide, glidant) is used.

Justification of Material Grade

The basis for selection of Aerosil 200 Pharma is reduced lot-to-lot variability reported by the qualified supplier, demonstrated across three pilot-scale batches.

Grade selection and control follow the principles of Ph. Eur. general monograph 2034.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

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Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

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Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Deviations observed during the campaign were investigated and closed with no impact to product quality.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Deviations observed during the campaign were investigated and closed with no impact to product quality. No changes to the approved analytical procedures are proposed in this submission. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Cleaning verification swab results were below the health-based exposure limits derived for the product. The control strategy links critical quality attributes to critical process parameters identified during development. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Deviations observed during the campaign were investigated and closed with no impact to product quality. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

The control strategy links critical quality attributes to critical process parameters identified during development. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection.

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The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. The control strategy links critical quality attributes to critical process parameters identified during development. No changes to the approved analytical procedures are proposed in this submission. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

No changes to the approved analytical procedures are proposed in this submission. Deviations observed during the campaign were investigated and closed with no impact to product quality. The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.

Commercial Control Strategy — Excipients

Component	Function	Grade per current specification
Colloidal silicon dioxide	glidant	Cab-O-Sil M-5P
Reference		SPEC-EX-8095 Rev 05

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

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Container Closure System

The commercial packaging configuration consists of high-density polyethylene bottles with child-resistant polypropylene closures and induction seal liners. Container closure integrity was demonstrated by dye ingress testing. The packaging components comply with applicable food-contact regulations and compendial requirements for plastic packaging systems.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. The control strategy links critical quality attributes to critical process parameters identified during development.